

Sheth L.U.J. College And Sir M.V. College of Science

Practical 5: Key Exchange using Diffie-Hellman

Aim:-Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

Code:-

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Practical5Dybergy - C:/Users/info/OneDrive/Desktop/Practical5Dybergy (3.14.6)
File Edit Format Run Options Window Help
import tkinter as tk
from tkinter import messagebox

def generate_key():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(alice_private.get())
        b = int(bob_private.get())

        alice_public = pow(g, a, p)
        bob_public = pow(g, b, p)

        alice_shared = pow(bob_public, a, p)
        bob_shared = pow(alice_public, b, p)

        alice_public_entry.delete(0, tk.END)
        alice_public_entry.insert(0, alice_public)

        bob_public_entry.delete(0, tk.END)
        bob_public_entry.insert(0, bob_public)

        alice_shared_entry.delete(0, tk.END)
        alice_shared_entry.insert(0, alice_shared)

        bob_shared_entry.delete(0, tk.END)
        bob_shared_entry.insert(0, bob_shared)

        if alice_shared == bob_shared:
            result_label.config(
                text="✓ KEY EXCHANGE SUCCESSFUL\nBoth Alice and Bob have the same shared secret key.",
                fg="green"
            )
        else:
            result_label.config(
                text="✗ KEY EXCHANGE FAILED",
                fg="red"
            )
    except ValueError:
        messagebox.showerror(
            "Error",
            "Please enter valid numbers."
        )

def clear_data():
    for entry in [
        p_entry,
        g_entry,
        alice_private,
        alice_public_entry,
        alice_shared_entry,
        bob_private,
        bob_public_entry,
        bob_shared_entry
    ]:
        entry.delete(0, tk.END)

    result_label.config(text="")

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("700x700")
root.resizable(True, True)

title = tk.Label(
    root,
    text="DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 16, "bold")
)
title.pack(pady=20)

subtitle = tk.Label(
    root,
    text="Secure Key Exchange Over an Insecure Network",
    font=("Arial", 11)
)
subtitle.pack(pady=5)

parameters = tk.LabelFrame(
    root,
    text="Public Parameters",
    font=("Arial", 12, "bold"),
    padx=20,
    pady=15
)
parameters.pack(padx=40, pady=15, fill="x")

tk.Label(
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tk.Label(
    parameters,
    text="Prime Number (p):",
    font=("Arial", 11)
).grid(row=0, column=0, padx=15, pady=8)

p_entry = tk.Entry(
    parameters,
    width=20,
    font=("Arial", 11)
)
p_entry.grid(row=0, column=1, padx=15, pady=8)
p_entry.insert(0, "23")

tk.Label(
    parameters,
    text="Generator (g):",
    font=("Arial", 11)
).grid(row=1, column=0, padx=15, pady=8)

g_entry = tk.Entry(
    parameters,
    width=20,
    font=("Arial", 11)
)
g_entry.grid(row=1, column=1, padx=15, pady=8)
g_entry.insert(0, "5")

alice_frame = tk.LabelFrame(
    root,
    text="ALICE",
    font=("Arial", 13, "bold"),
    padx=20,
    pady=15
)
alice_frame.pack(side="left", padx=40, pady=10)

tk.Label(
    alice_frame,
    text="Private Key:",
    font=("Arial", 11)
).grid(row=0, column=0, padx=10, pady=10)

alice_private = tk.Entry(
    alice_frame,
    width=15,
    font=("Arial", 11)
)
alice_private.grid(row=0, column=1, padx=10, pady=10)
alice_private.insert(0, "6")

tk.Label(
    alice_frame,
    text="Public Key:",
    font=("Arial", 11)
).grid(row=1, column=0, padx=10, pady=10)

alice_public_entry = tk.Entry(
    alice_frame,
    width=15,
    font=("Arial", 11)
)
alice_public_entry.grid(row=1, column=1, padx=10, pady=10)

tk.Label(
    alice_frame,
    text="Shared Secret:",
    font=("Arial", 11)
).grid(row=2, column=0, padx=10, pady=10)

alice_shared_entry = tk.Entry(
    alice_frame,
    width=15,
    font=("Arial", 11)
)
alice_shared_entry.grid(row=2, column=1, padx=10, pady=10)

bob_frame = tk.LabelFrame(
    root,
    text="BOB",
    font=("Arial", 13, "bold"),
    padx=20,
    pady=15
)
bob_frame.pack(side="left", padx=40, pady=10)
```

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```
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root,
text="BOB",
font=("Arial", 13, "bold"),
padx=20,
pady=15
)
bob_frame.pack(side="right", padx=40, pady=10)

tk.Label(
    bob_frame,
    text="Private Key:",
    font=("Arial", 11)
).grid(row=0, column=0, padx=10, pady=10)

bob_private = tk.Entry(
    bob_frame,
    width=15,
    font=("Arial", 11)
)
bob_private.grid(row=0, column=1, padx=10, pady=10)
bob_private.insert(0, "15")

tk.Label(
    bob_frame,
    text="Public Key:",
    font=("Arial", 11)
).grid(row=1, column=0, padx=10, pady=10)

bob_public_entry = tk.Entry(
    bob_frame,
    width=15,
    font=("Arial", 11)
)
bob_public_entry.grid(row=1, column=1, padx=10, pady=10)

tk.Label(
    bob_frame,
    text="Shared Secret:",
    font=("Arial", 11)
).grid(row=2, column=0, padx=10, pady=10)

bob_shared_entry = tk.Entry(
    bob_frame,
    width=15,
    font=("Arial", 11)
)

text="Shared Secret:",
font=("Arial", 11)
).grid(row=2, column=0, padx=10, pady=10)

bob_shared_entry = tk.Entry(
    bob_frame,
    width=15,
    font=("Arial", 11)
)
bob_shared_entry.grid(row=2, column=1, padx=10, pady=10)

button_frame = tk.Frame(root)
button_frame.pack(pady=30)

generate_button = tk.Button(
    button_frame,
    text="GENERATE SHARED KEY",
    command=generate_key,
    font=("Arial", 12, "bold"),
    width=22,
    height=2
)
generate_button.grid(row=0, column=0, padx=10)

clear_button = tk.Button(
    button_frame,
    text="CLEAR",
    command=clear_data,
    font=("Arial", 12, "bold"),
    width=12,
    height=2
)
clear_button.grid(row=0, column=1, padx=10)

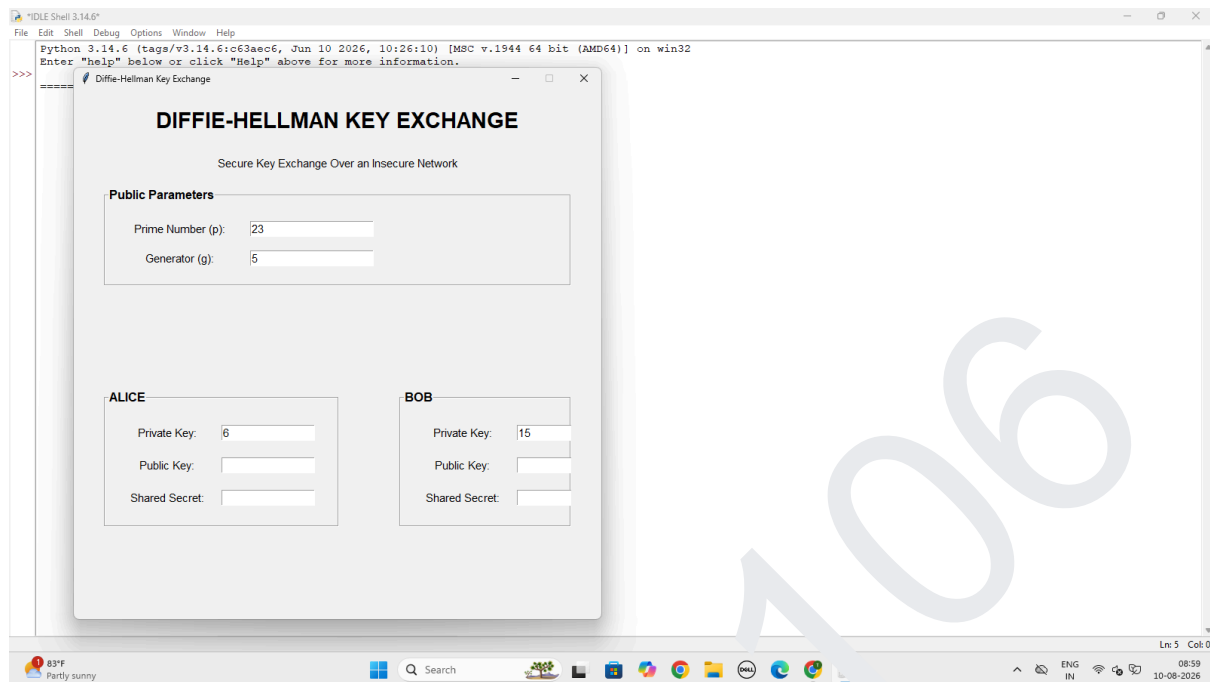
result_label = tk.Label(
    root,
    text="",
    font=("Arial", 12, "bold"),
    justify="center"
)
result_label.pack(pady=10)

root.mainloop()
```

Output:-

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